Project: Pandas Analyzer & Data Visualization

Objective:

Develop a comprehensive Sales Data Analysis and Visualization tool in Python. This project will utilize Pandas for data manipulation and analysis, Matplotlib and Seaborn for data visualization. Students will gain hands-on experience in handling real-world sales datasets, performing various data operations, and creating insightful visualizations to support business decisions.

Project Requirements:

Data Acquisition:

- Obtain a sales dataset in CSV format. (You can use publicly available datasets such as the Sales Dataset from Kaggle or create a synthetic dataset with columns like Date, Product, Sales, Region, Profit, etc.)

Environment Setup:

- Use Jupyter Notebook for developing and documenting the project.
- Install necessary libraries: `pip install pandas matplotlib seaborn`

Class Structure:

- **SalesDataAnalyzer** class encapsulating all functionalities.
- **Attributes:**
  - self.data: Pandas DataFrame to store sales data.
- **Methods:**
  - __init__: Constructor to load data.
  - __del__: Destructor to perform any cleanup if necessary.
  - load_data(file_path): Load data from a CSV file.
  - explore_data(): Display basic information about the dataset.
  - clean_data(): Handle missing values and perform data cleaning.
  - mathematical_operations(): Perform mathematical operations on data.
  - combine_data(other_dataframe): Combine current DataFrame with another.
  - split_data(): Split DataFrame into multiple DataFrames based on criteria.
  - search_sort_filter(): Implement search, sort, and filter functionalities.

- aggregate_functions(): Apply aggregating functions like sum, mean, etc.
- statistical_analysis(): Perform statistical computations.
- create_pivot_table(): Generate pivot tables for data summarization.
- visualize_data(): Create various plots using Matplotlib and Seaborn.
- Additional helper methods as needed.

Functional Requirements:

A. Creation, Indexing, and Slicing of Numpy Arrays:

- Convert relevant DataFrame columns to Numpy arrays.
- Demonstrate indexing and slicing operations on these arrays.

B. Mathematical Operations, Combining and Splitting Arrays:

- Perform element-wise mathematical operations.
- Combine DataFrames using concat, merge, or join.
- Split DataFrames based on regions, products, or other criteria.

C. Search, Sort, Filter Arrays, Aggregating Functions:

- Search for specific sales figures or products.
- Sort data based on sales, profit, or other metrics.
- Filter data for specific regions or time periods.
- Apply aggregating functions like sum(), mean(), count().

D. Statistical Functions in Arrays:

- Calculate statistics such as standard deviation, variance, percentiles.
- Use functions like describe(), std(), var(), quantile().

E. Data Visualization with Matplotlib:

- Create Bar, Line, Scatter, Pie, and Histogram plots.
- Implement Stack Plot Chart.
- Add legends and titles to plots.
- Create subplots to display multiple plots in a single figure.
- Save visualizations as image files (e.g., PNG, JPEG).

F. Data Visualization with Seaborn:

- Create enhanced visualizations like heatmaps, box plots, etc.
- Customize plots with Seaborn's styling options.

Object-Oriented Programming (OOP):

- Encapsulate data and functionalities within the SalesDataAnalyzer class.

- Demonstrate inheritance if extending functionalities (optional).
- Use super() to call parent class methods if inheritance is implemented.
- Override methods if necessary.
- Utilize operator overloading for custom behaviors (optional).

User Interface (UI):

- Create a menu-driven interface to let users choose from different options.
- Allow users to exit the program from the main menu.

Program Flow:

Initialization:

- Instantiate the SalesDataAnalyzer class.
- Load the sales data from a CSV file.
- Handle any exceptions during file loading.

Data Exploration and Cleaning:

- Explore the dataset using head(), info(), describe().
- Identify and handle missing data.
- Perform necessary data type conversions.

Data Manipulation:

- Perform mathematical operations on sales data.
- Combine multiple DataFrames if multiple data sources are provided.
- Split data based on predefined criteria (e.g., by region or product).

Data Analysis:

- Search for specific records or patterns.
- Sort and filter data based on user inputs.
- Apply aggregating and statistical functions to derive insights.

Data Visualization:

- Generate various plots to visualize sales trends, distributions, and comparisons.
- Customize plots with titles, labels, legends.
- Save important visualizations to files for reporting.

Advanced Operations:

- Create pivot tables to summarize sales data.
- Re-index and alter labels for better data representation.
- Use groupby() and transform() for complex data aggregations.

Conclusion:

- Summarize the findings and visualizations.
- Provide options to export analysis results or visualizations.

Example Console Interaction:

```
========== Data Analysis & Visualization Program ==========
Please select an option:
1. Load Dataset
2. Explore Data
3. Perform DataFrame Operations
4. Handle Missing Data
5. Generate Descriptive Statistics
6. Data Visualization
7. Save Visualization
8. Exit
============================================================

Enter your choice: 1

== Load Dataset ==
Enter the path of the dataset (CSV file): data/sales_data.csv
Dataset loaded successfully!

========== Data Analysis & Visualization Program ==========
Please select an option:
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============================================================

Enter your choice: 2

== Explore Data ==
1. Display the first 5 rows
```

```
2. Display the last 5 rows
3. Display column names
4. Display data types
5. Display basic info
Enter your choice: 1

   SalesID     Product      Region  Sales  Year
0       101  Product A     North    500   2022
1       102  Product B      East    600   2022
2       103  Product C  West Coast   700   2022
3       104  Product D      South    800   2022
4       105  Product E     Central   550   2022

========== Data Analysis & Visualization Program ==========
Please select an option:
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============================================================

Enter your choice: 4

== Handle Missing Data ==
1. Display rows with missing values
2. Fill missing values with mean
3. Drop rows with missing values
4. Replace missing values with a specific value
Enter your choice: 1

No missing values found in the dataset!

========== Data Analysis & Visualization Program ==========
Please select an option:
1. Load Dataset
2. Explore Data
3. Perform DataFrame Operations
4. Handle Missing Data
5. Generate Descriptive Statistics
```

```
6. Data Visualization
7. Save Visualization
8. Exit
============================================================

Enter your choice: 6

== Data Visualization ==
1. Bar Plot
2. Line Plot
3. Scatter Plot
4. Pie Chart
5. Histogram
6. Stack Plot
Enter your choice: 3

== Scatter Plot ==
Enter x-axis column name: Sales
Enter y-axis column name: Year
Generating scatter plot...
Scatter plot displayed successfully!

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Please select an option:
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8. Exit
============================================================

Enter your choice: 7

== Save Visualization ==
Enter file name to save the plot (e.g., scatter_plot.png): scatter_plot.png
Visualization saved as scatter_plot.png successfully!

========== Data Analysis & Visualization Program ==========
Please select an option:
1. Load Dataset
```

```
2. Explore Data
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5. Generate Descriptive Statistics
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8. Exit
============================================================

Enter your choice: 8

Exiting the program. Goodbye!
```

Instructions:

Task Completion: Ensure that you attempt all the assigned tasks given to you as part of the exam. Complete each task to the best of your ability, following the instructions provided.

Assumptions: Make suitable assumptions wherever necessary, based on the requirements and instructions provided. Document any assumptions made in your project documentation or README.md file.

GitHub Repository: Create a GitHub repository to host your project. Upload your project files, including source code, and documentation to the repository. Ensure that you provide a clear and descriptive README.md file.

No Copying: Do not copy code or any other content from your classmates or any other source. Plagiarism is strictly prohibited and can result in severe consequences, including academic penalties. Ensure that all the code and content in your project are original and properly attributed to the appropriate sources, if applicable.

Submission: Once you have completed your project and uploaded it to your GitHub repository, submit the GitHub repository link to your instructor or as instructed. Double-check that your repository is properly organized and includes all the required files as per the instructions provided.

Remember to follow the instructions provided professionally, make suitable assumptions wherever necessary, and avoid copying code or content from unauthorized sources. Good luck with your project work!